

Discovering Computers 2016

Tools, Apps, Devices, and the Impact of Technology

Chapter 10

Communicating Digital Content



Objectives Overview

Discuss the purpose of the components required for successful communications and identify various sending and receiving devices

Differentiate among LANs, MANs, WANs, and PANs

Differentiate between client/server and peer-to-peer networks

Explain the purpose of communications software

Describe the various network communications standards and protocols

Objectives Overview

Describe various
types of
communications
lines

Describe
commonly used
communications
devices

Discuss different
ways to set up and
configure a home
network

Differentiate
among physical
transmission
media

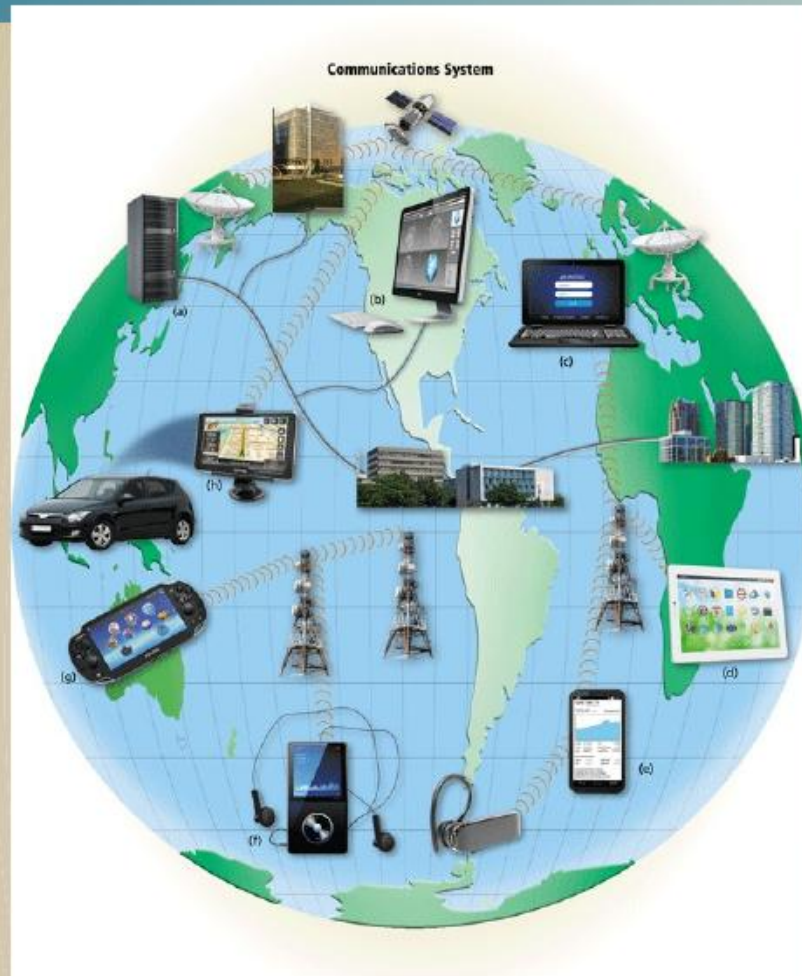
Differentiate
among wireless
transmission
media

Communications

- Digital communications describes a process in which two or more computers or devices transfer data, instructions, and information



Communications



Networks

- A **network** is a collection of computers and devices connected together via communications devices and transmission media
- Advantages of a network include:

Facilitating
communications

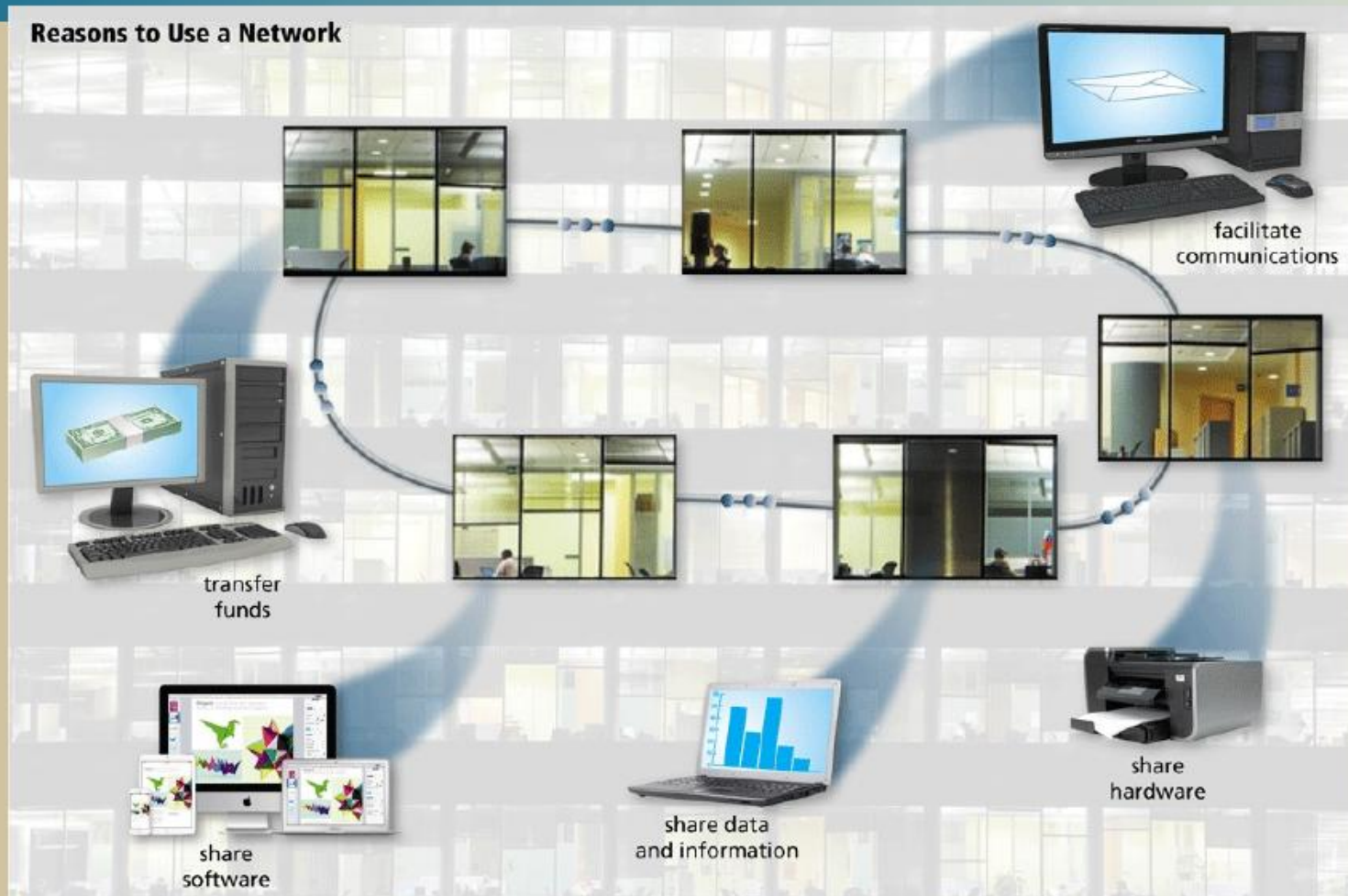
Sharing
hardware

Sharing data and
information

Sharing software

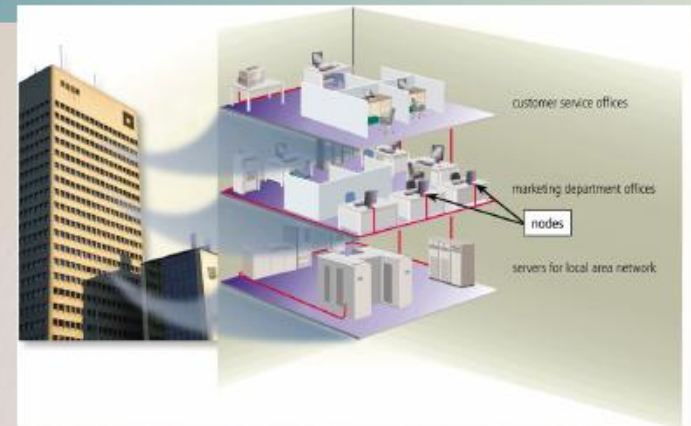
Transferring
funds

Networks



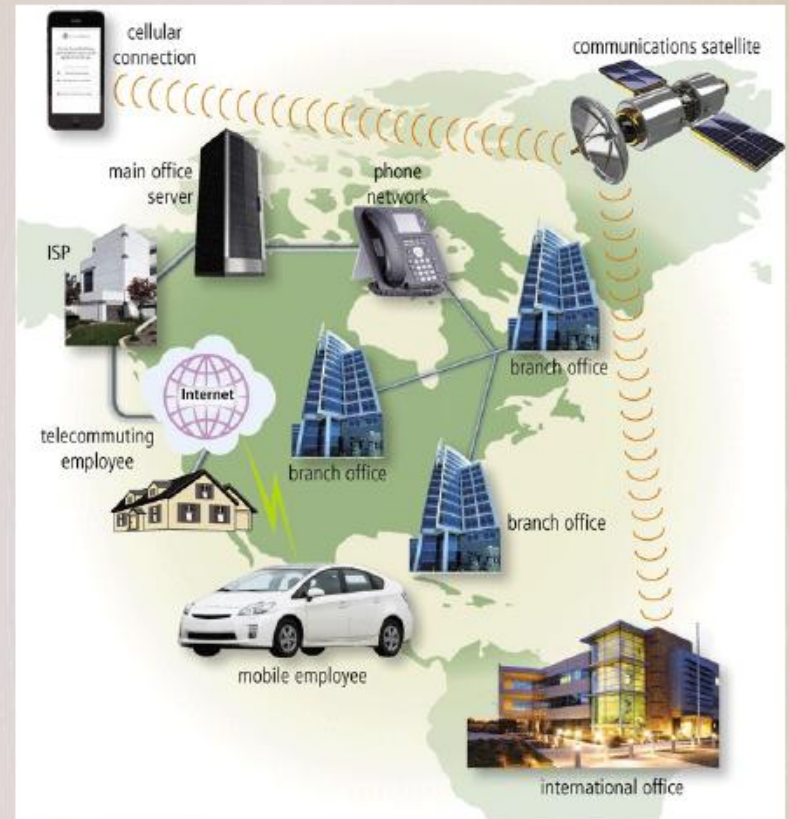
Networks

- A **local area network (LAN)** is a network that connects computers and devices in a limited geographical area
- A **wireless LAN (WLAN)** is a LAN that uses no physical wires



Networks

- A metropolitan area network (MAN) connects LANs in a metropolitan area
- A **wide area network (WAN)** is a network that covers a large geographic area
- A **personal area network (PAN)** is a network that connects computers and devices in an individual's workspace with wired and wireless technology



Networks

- The configuration of computers, devices, and media on a network is sometimes called the network architecture

Client/server network



Peer-to-peer network



Communications Software

- **Communications software** consists of programs and apps that:

Help users establish a connection to another computer, mobile device, or network

Manage the transmission of data, instructions, and information

Provide an interface for users to communicate with one another

Network Communications Standards and Protocols

Ethernet

Token
ring

TCP/IP

Wi-Fi

Bluetooth

UWB

IrDA

RFID

NFC

LTE

Network Communications Standards and Protocols

Ethernet is a network standard that specifies no central computer or device on the network (nodes) should control when data can be transmitted

The **token ring** standard specifies that computers and devices on the network share or pass a special signal (token)

TCP/IP is a network protocol that defines how messages (data) are routed from one end of a network to another

Network Communications Standards and Protocols

How Communications Standards Might Work Together



Step 2a: TCP/IP

Your computer uses the TCP/IP standard to establish a connection with the web server that stores the requested webpage, divide the webpage into packets, provide an address for each packet, and reassemble the webpage once it arrives at your computer. Routers send the packets over the Internet from the web server to your computer.

Step 2b: ETHERNET

The Ethernet standard controls how devices (adapter cards, routers, modems, etc.) share access to the media (cables and lines) and how devices transmit data over the transmission media.



Network Communications Standards and Protocols

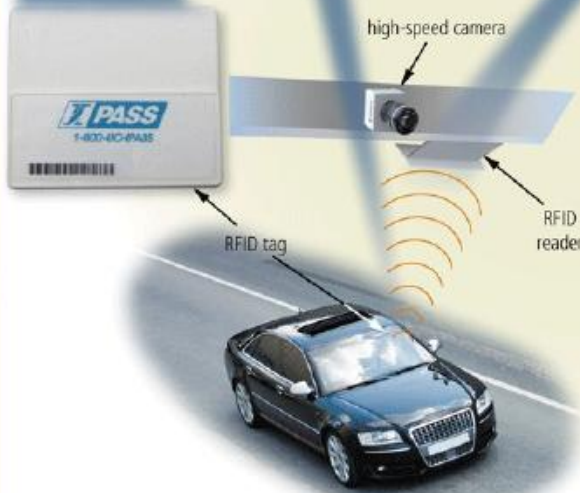
- **Wi-Fi** identifies any network based on the **802.11** standard that specifies how two wireless devices communicate over the air with each other
- **LTE** is a network standard that defines how high-speed cellular transmissions use broadcast radio to transmit data for mobile communications
- **Bluetooth** is a network protocol that defines how two Bluetooth devices use short-range radio waves to transmit data
- **UWB (ultra-wideband)** is a network standard that specifies how two UWB devices use short-range radio waves to communicate at high speeds with each other
- **IrDA** transmits data wirelessly via infrared (IR) light waves
- **RFID** is a protocol that defines how a network uses radio signals to communicate with a tag placed in or attached to an object, an animal, or a person

Network Communications Standards and Protocols

How Electronic RFID Toll Collection Works

Step 1

Motorist purchases an RFID transponder or RFID tag and attaches it to the vehicle's windshield.



Step 2

As the vehicle approaches the tollbooth, the RFID reader in the tollbooth sends a radio wave that activates the windshield-mounted RFID tag. The activated tag sends vehicle information to the RFID reader.



Step 3

The RFID reader sends the vehicle information to the lane controller. The lane controller, which is part of a local area network, transmits the vehicle information to a central computer that subtracts the toll from the motorist's account. If the vehicle does not have an RFID tag, a high-speed camera takes a picture of the license plate and the computer prints a violation notice, which is mailed to the motorist.



Network Communications Standards and Protocols

- **NFC** (near field communication) is a protocol, based on RFID, that defines how a network uses close-range radio signals to communicate between two devices or objects equipped with NFC technology



Communications Lines

Dedicated line

Cable

DSL

ISDN

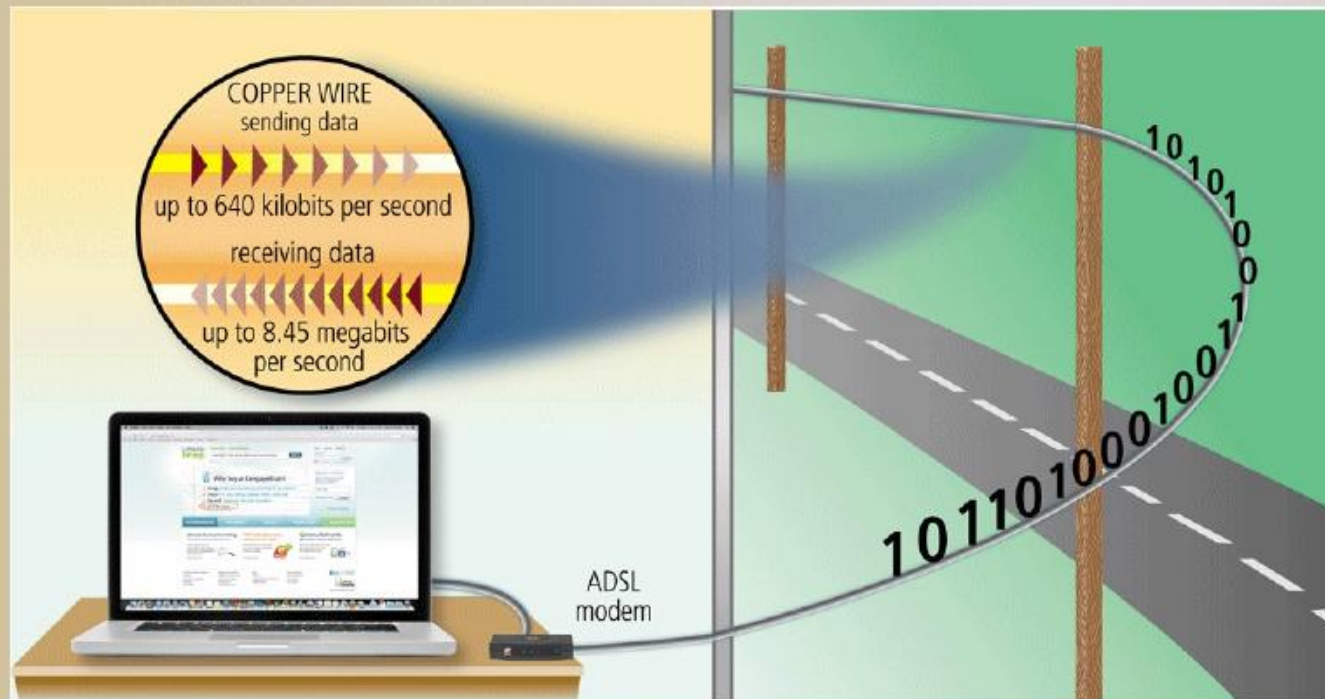
Communications Lines

Table 10-2 Speeds of Various Dedicated Digital Lines

Type of Line	Transfer Rates
Cable	256 Kbps to 52 Mbps
DSL	256 Kbps to 8.45 Mbps
ISDN	Up to 1.54 Mbps
FTTP	5 Mbps to 300 Mbps
Fractional T1	128 Kbps to 768 Kbps
T1	1.544 Mbps
T3	44.736 Mbps
ATM	155 Mbps to 622 Mbps, can reach 10 Gbps

Communications Lines

- ADSL is a type of DSL that supports faster downstream rates than upstream rates



Communications Devices

- A **communications device** is any type of hardware capable of transmitting data, instructions, and information between a sending device and a receiving device

Communications Devices

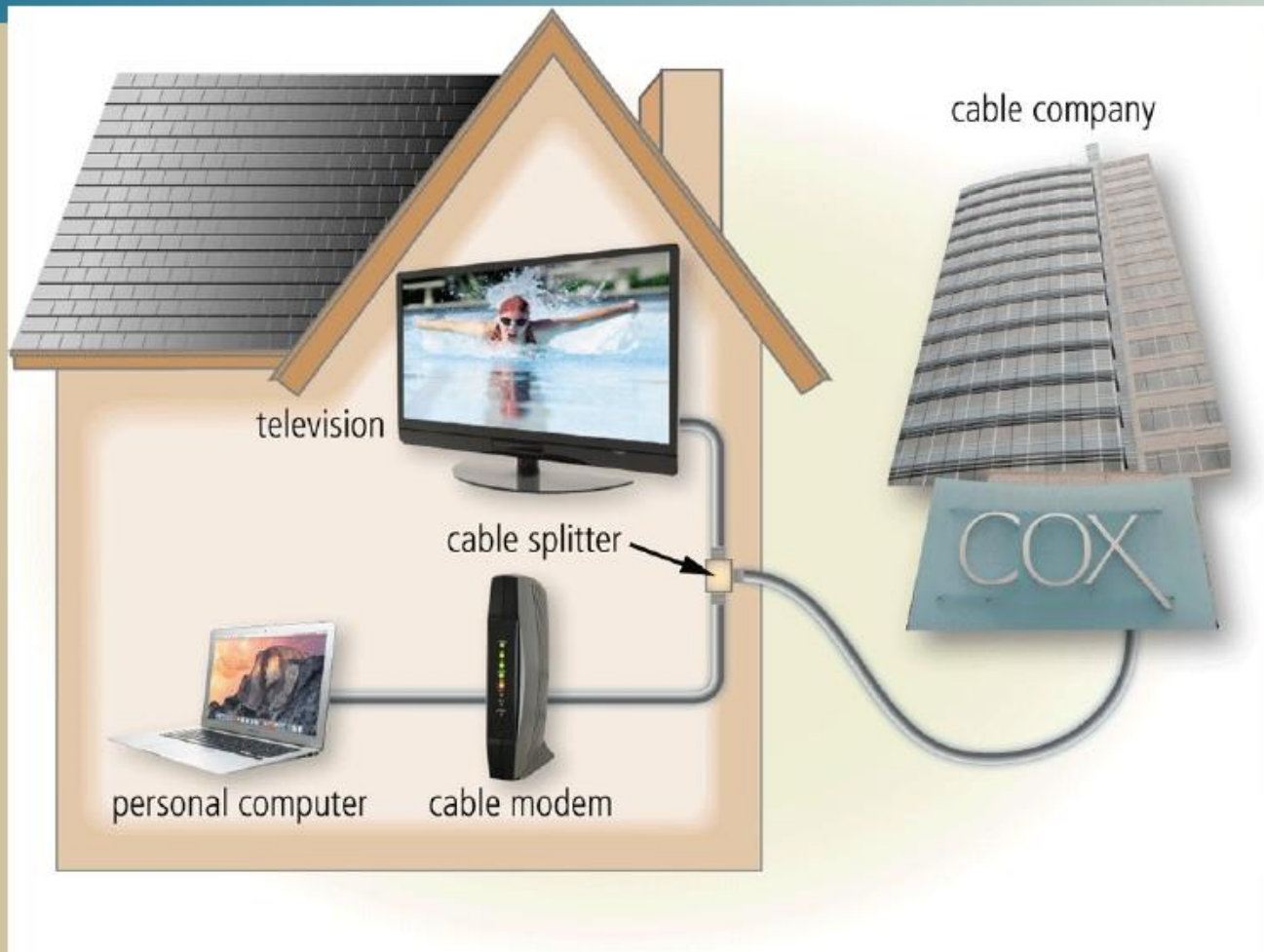
- A broadband modem sends and receives data and information to and from a digital line



Cable modem

DSL modem

Communications Devices



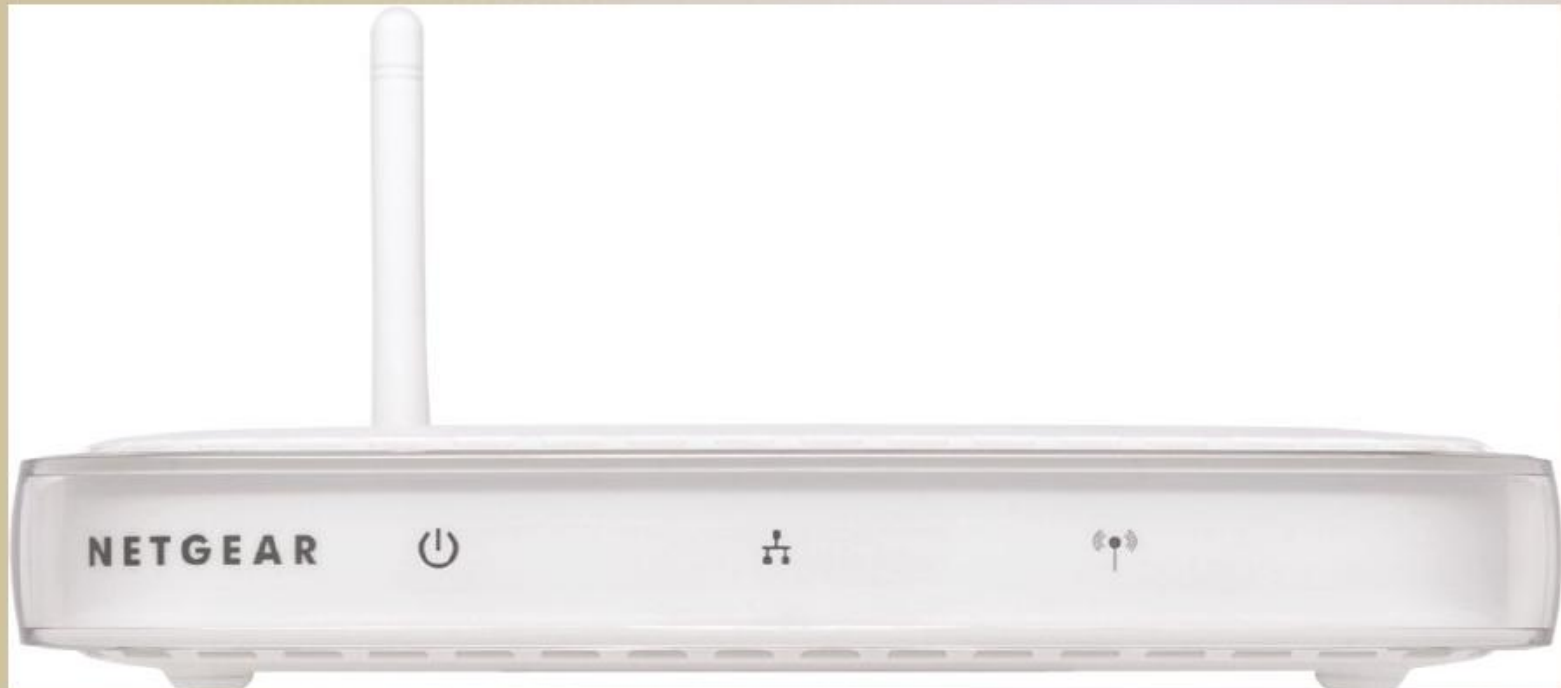
Communications Devices

- A wireless modem uses a mobile phone provider's network to connect to the Internet wirelessly from a computer or mobile device



Communications Devices

- A wireless access point (WAP) is a central communications device that allows computers and devices to transfer data wirelessly among themselves or to a wired network



Communications Devices

- A router connects multiple computers or other routers together and transmits data to its correct destination on a network



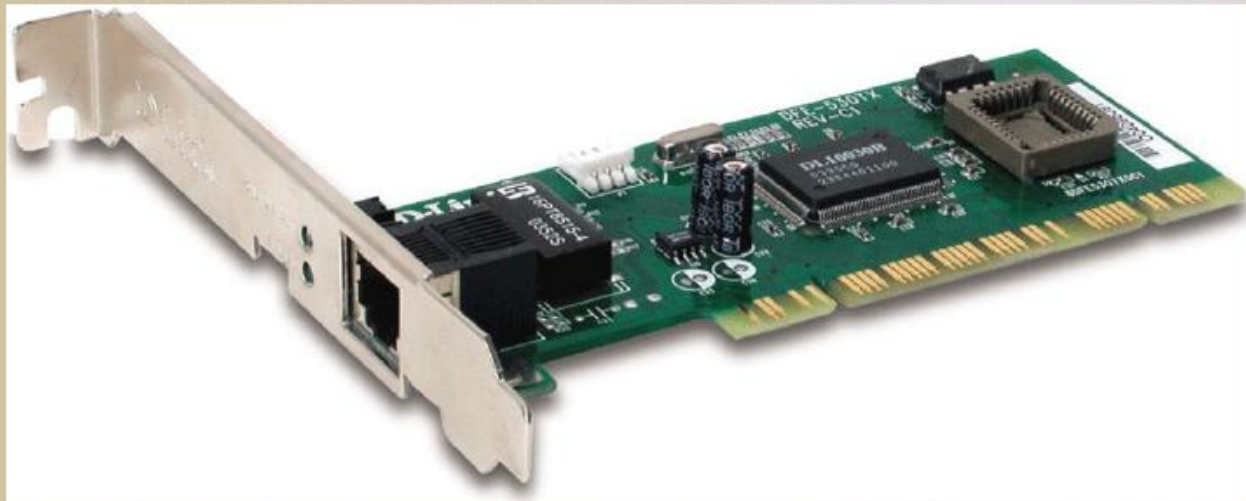
Communications Devices

- Some routers provide additional functionality:
 - Wireless router
 - Broadband router
 - Broadband wireless router
 - Mobile broadband wireless router



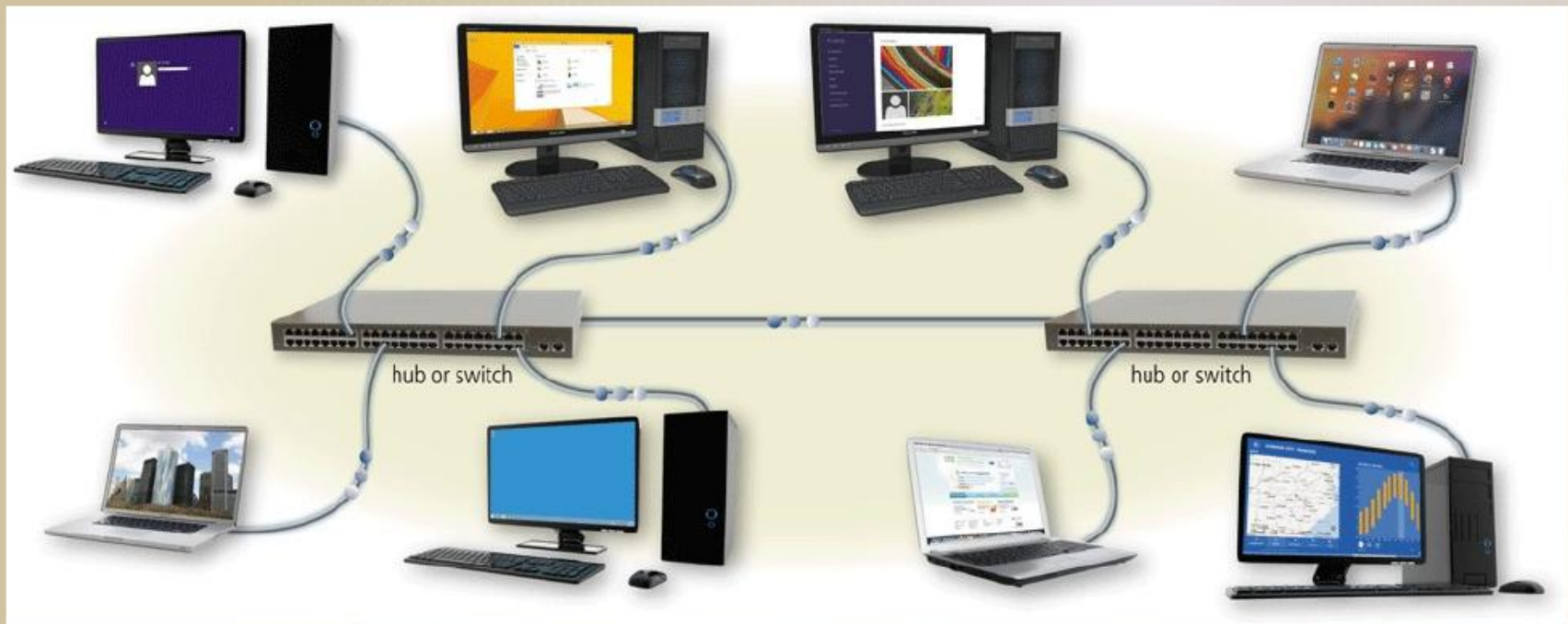
Communications Devices

- A network card enables a computer or device that does not have built-in networking capability to access a network
- Available in a variety of styles



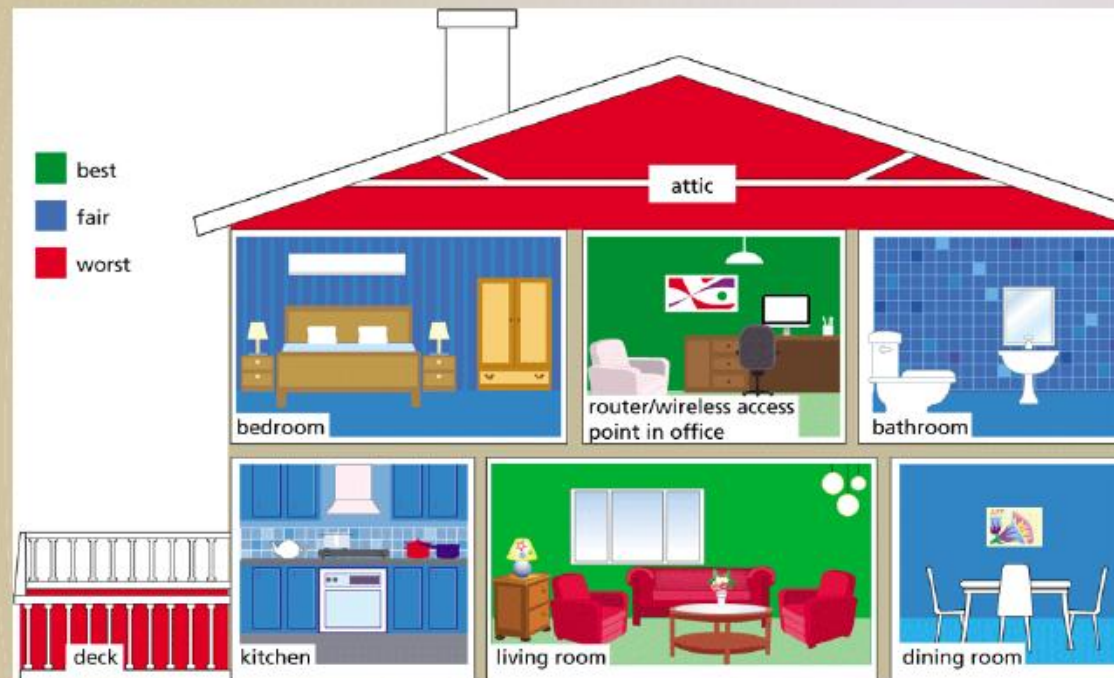
Communications Devices

- A hub or switch is a device that provides a central point for cables in a network



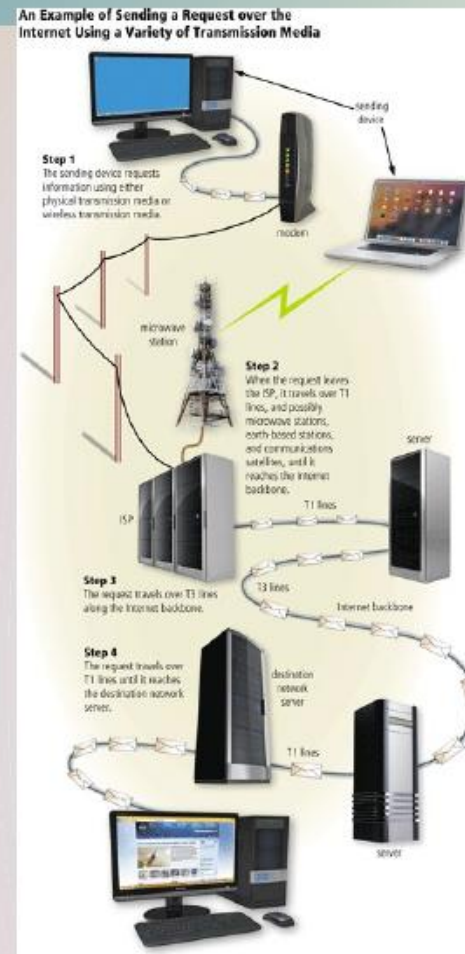
Home Networks

- Many home users connect multiple computers and devices together in a **home network**



Transmission Media

- Transmission media carries one or more communications signals
- **Broadband** media transmit multiple signals simultaneously
- The amount of data, instructions, and information that can travel over transmission media sometimes is called the **bandwidth**
- **Latency** is the time it takes a signal to travel from one location to another on a network



Physical Transmission Media

Table 10-3 Transfer Rates for Physical Transmission Media Used in LANs

Type of Cable and LAN	Maximum Transfer Rate
-----------------------	-----------------------

Twisted-Pair Cable

- | | |
|---------------------------------|-------------------|
| • 10Base-T (Ethernet) | 10 Mbps |
| • 100Base-T (Fast Ethernet) | 100 Mbps |
| • 1000Base-T (Gigabit Ethernet) | 1 Gbps |
| • Token ring | 4 Mbps to 16 Mbps |

Coaxial Cable

- | | |
|--------------------------------|---------|
| • 10Base2 (ThinWire Ethernet) | 10 Mbps |
| • 10Base5 (ThickWire Ethernet) | 10 Mbps |

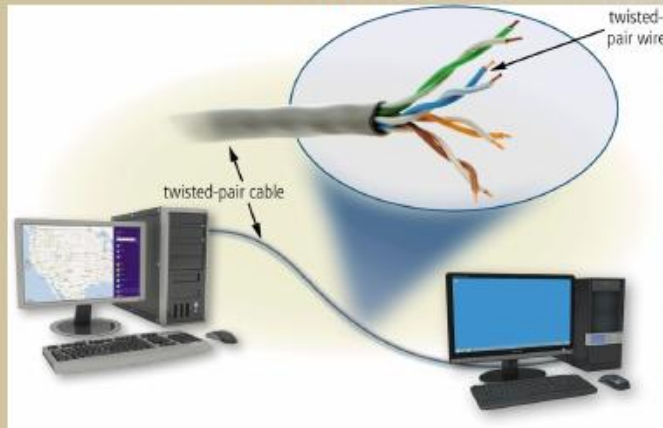
Fiber-Optic Cable

- | | |
|--|----------|
| • 10Base-F (Ethernet) | 10 Mbps |
| • 100Base-FX (Fast Ethernet) | 100 Mbps |
| • FDDI (Fiber Distributed Data Interface) token ring | 100 Mbps |
| • Gigabit Ethernet | 1 Gbps |
| • 10-Gigabit Ethernet | 10 Gbps |
| • 40-Gigabit Ethernet | 40 Gbps |
| • 100-Gigabit Ethernet | 100 Gbps |

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Physical Transmission Media

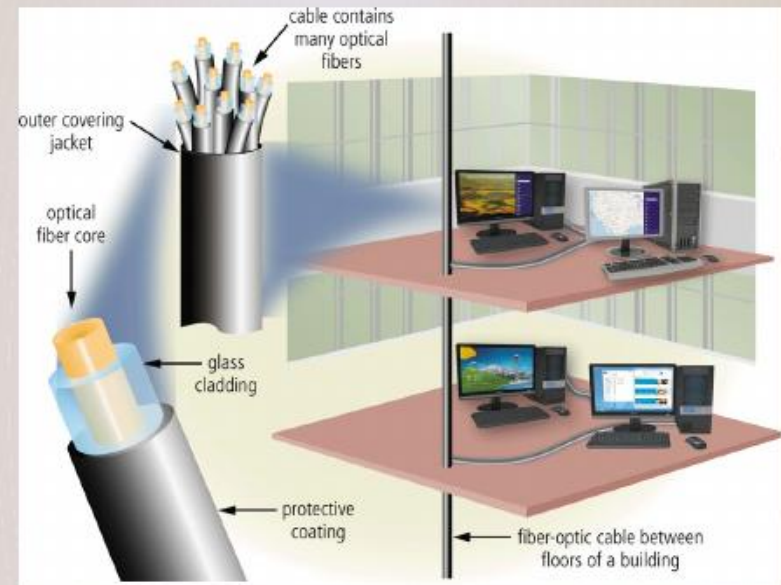
Twisted-pair cable



Coaxial cable



Fiber-optic cable



Wireless Transmission Media

Table 10-4 Wireless Transmission Media Transfer Rates

Medium		Maximum Transfer Transmission Rate
Infrared		115 Kbps to 4 Mbps
Broadcast radio	• Bluetooth	1 Mbps to 24 Mbps
	• 802.11b	11 Mbps
	• 802.11a	54 Mbps
	• 802.11g	54 Mbps
	• 802.11n	300 Mbps
	• 802.11ac	500 Mbps to 1 Gbps
	• 802.11ad	up to 7 Gbps
	• UWB	110 Mbps to 480 Mbps
Cellular radio	• 2G	9.6 Kbps to 144 Kbps
	• 3G	144 Kbps to 3.84 Mbps
	• 4G	Up to 100 Mbps
Microwave radio		Up to 10 Gbps
Communications satellite		Up to 2.56 Tbps

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Wireless Transmission Media

- **Broadcast radio** is a wireless transmission medium that distributes radio signals through the air over long distances
- **Cellular radio** is a form of broadcast radio that is in wide use for mobile communications



Wireless Transmission Media

- **Microwaves** are radio waves that provide a high-speed signal transmission



Wireless Transmission Media

- A **communications satellite** is a space station that receives microwave signals from an earth-based station, amplifies it, and broadcasts the signal over a wide area to any number of earth-based stations



Wireless Transmission Media

- A **GPS** (global positioning system) is a navigation system that consists of one or more earth-based receivers that accept and analyze signals sent by satellites in order to determine the receiver's geographical location

Wireless Transmission Media



Summary

Networks and communications technologies

Various types of network architectures and standards and protocols

Communications software

Several types of communications lines and communications devices

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Chapter 10

Communicating Digital Content

Chapter 10 Complete

